

7. **Interoperability:** Networks all systems together and ensures interoperability of all data

According to the estimates of several experts, the IoT will consist of around 30 billion objects by 2020.

Organisations that adopted proprietary IoT 'platforms' are now finding themselves tied to limited functionality, locked into a particular vendor, and are hence looking at other options. Many of them are now looking at open source software as an alternative, recognising the value of the community as a hub for innovation and the continuous development for IoT.

In the IDC Global IoT Decision Maker Survey, August 2016, 64 per cent of the participants think open source is very or extremely important in their deployment of IoT. According to the World Wide Web Consortium ([w3.org](http://www.w3.org)), 91 per cent of IoT developers use open source software, open hardware, or open data in at least one part of their development stack.

In the coming years, the IoT market will move towards adopting an open source and open standards ecosystem that enables collaborative innovation, and promotes interoperability, flexibility and modularity.

Why open source makes sense for IoT

Today, IoT promotes the adoption of different open source technologies, standards and protocols that help devices communicate with one another. The following are the drivers that prompt organisations to adopt open source technologies for IoT.

1. **Cost:** Adoption of open source IoT frameworks involves no costs at all, as these are free for use. This encourages the community and organisations to implement IoT without any hesitation.
2. **Innovation:** Open source code from the community helps in building newer applications, leading to more innovation and agility. The developers are able to build different products, which will be interoperable across different OSs such as Android, Windows, iOS and Linux.
3. **Open APIs:** Use of open source APIs for the IoT framework offers a common gateway for different software, hardware and the systems to communicate with one another.
4. **Libraries:** An open source IoT framework offers a wide range of libraries, SDKs and open source hardware like Raspberry Pi and Arduino, ensuring that companies remain on the cutting-edge of technology by using different open sourced tools to customise IoT platforms.
5. **Security:** Open source software can protect individuals' data by implementing really strong encryption for the use of the general public (SSH, SSL, PGP, etc), and hence supply the building blocks for mobile security and the protection of data.

6. **Interoperability:** Adoption of open source solves the problem of interoperability.

Open source IoT architecture

There is a need for organisations to provide a validated, modular, flexible IoT architecture that is built to be open, interoperable and cost effective. The architecture should deliver end-to-end open source IoT that addresses enterprise level IoT needs.

The characteristics of open source IoT architecture are:

- Loosely coupled, modular and secure
- Platform independent
- Scalable, flexible and can be deployed anywhere
- Based on open standards
- Streaming analytics and machine learning
- Open and interoperable on the hybrid cloud
- Application agility and integration
- No vendor lock-in, since there are no rigid architectures or proprietary formats and components

Figure 1 depicts the IoT logical reference architecture as three tiers — edge, platform and enterprise. These tiers process the data flows and control flows based on usage activities across the enterprise systems. They span three networks.

1. **Edge:** This consists of IoT devices and the IoT gateway. The architectural characteristics of this tier, including its breadth of distribution and location, depend on the specific use cases of the enterprise. It is common for IoT devices to communicate using a relatively short range and specialised proximity network, due to power and processing limitations. The IoT gateway contains a data store for IoT device data, one or more services to analyse data streaming from the IoT devices or from the data store, and control applications. This gateway provides endpoints for device connectivity, facilitating bi-directional communication with the enterprise systems. It also implements edge intelligence with different levels of processing capabilities.
2. **Platform:** This receives, processes and forwards control commands from the enterprise tier to the edge tier. The platform tier consolidates, processes and analyses data flows from the edge tier, and provides management functions for devices and assets. It also offers non-domain-specific services such as data operations and analytics.
3. **Enterprise:** This tier receives data flows from the edge and platform tiers, and issues control commands to them. The enterprise tier implements enterprise domain-specific applications and decision support systems, and provides interfaces to end users, including operations.

The architecture enables bi-directional communication with devices via the intelligent-edge IoT gateways. Data is routed through the IoT integration hub for application integration within the enterprise applications and for aggregation into the centralised data management platform for deep analysis and machine learning.